

Activity Trends and User Behavior Insights **from Fitbit Data**

Data-Driven Insights from Daily Activity Tracking
2025, June 9th

Business Questions



What are users' activity patterns over time?



How do activity levels relate to calorie consumption?



Are there differences between active and non-active users?

Summary of Key Findings



Weekdays are more active than weekends in terms of steps.



Calorie consumption is stable regardless of step counts.

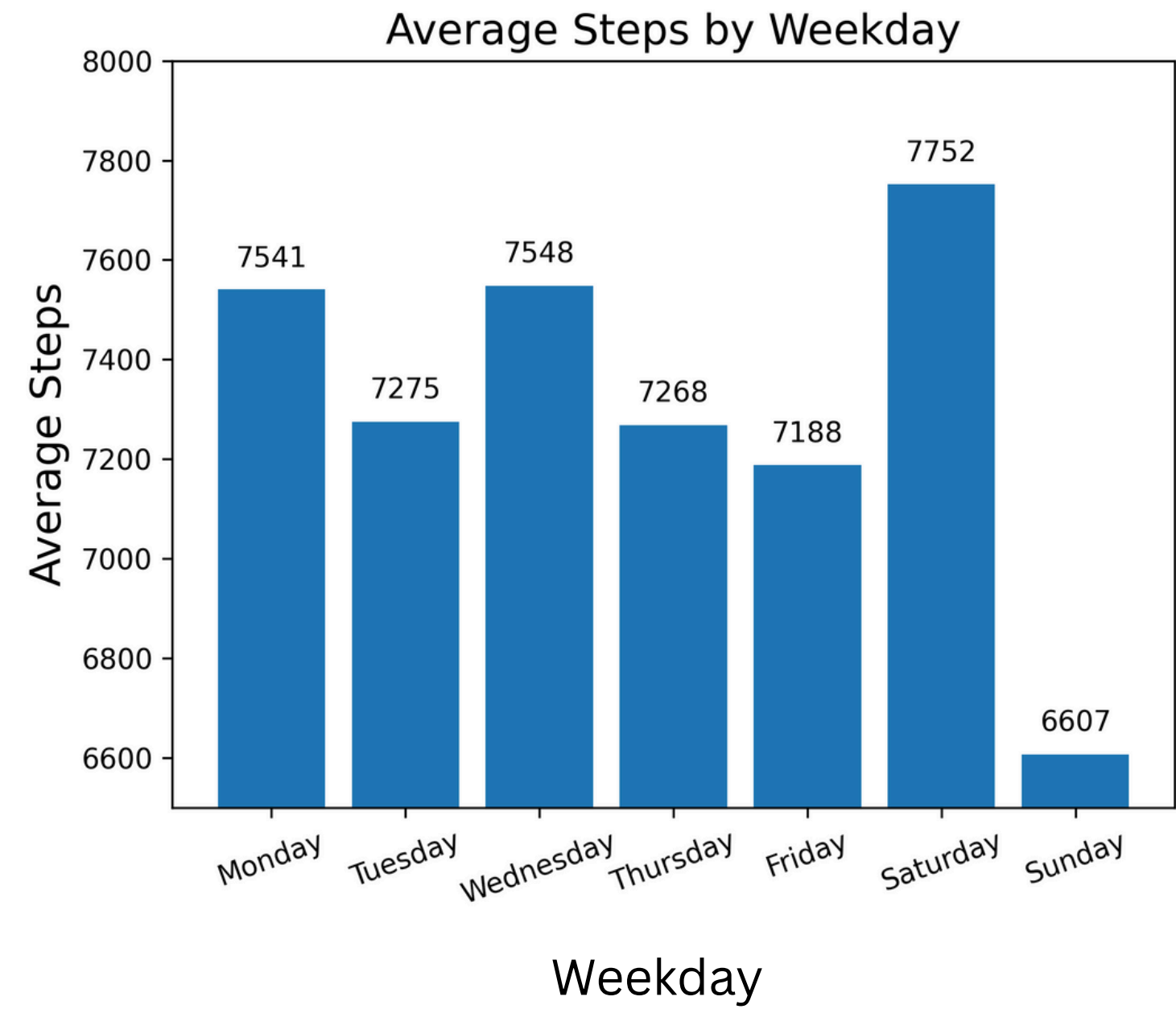
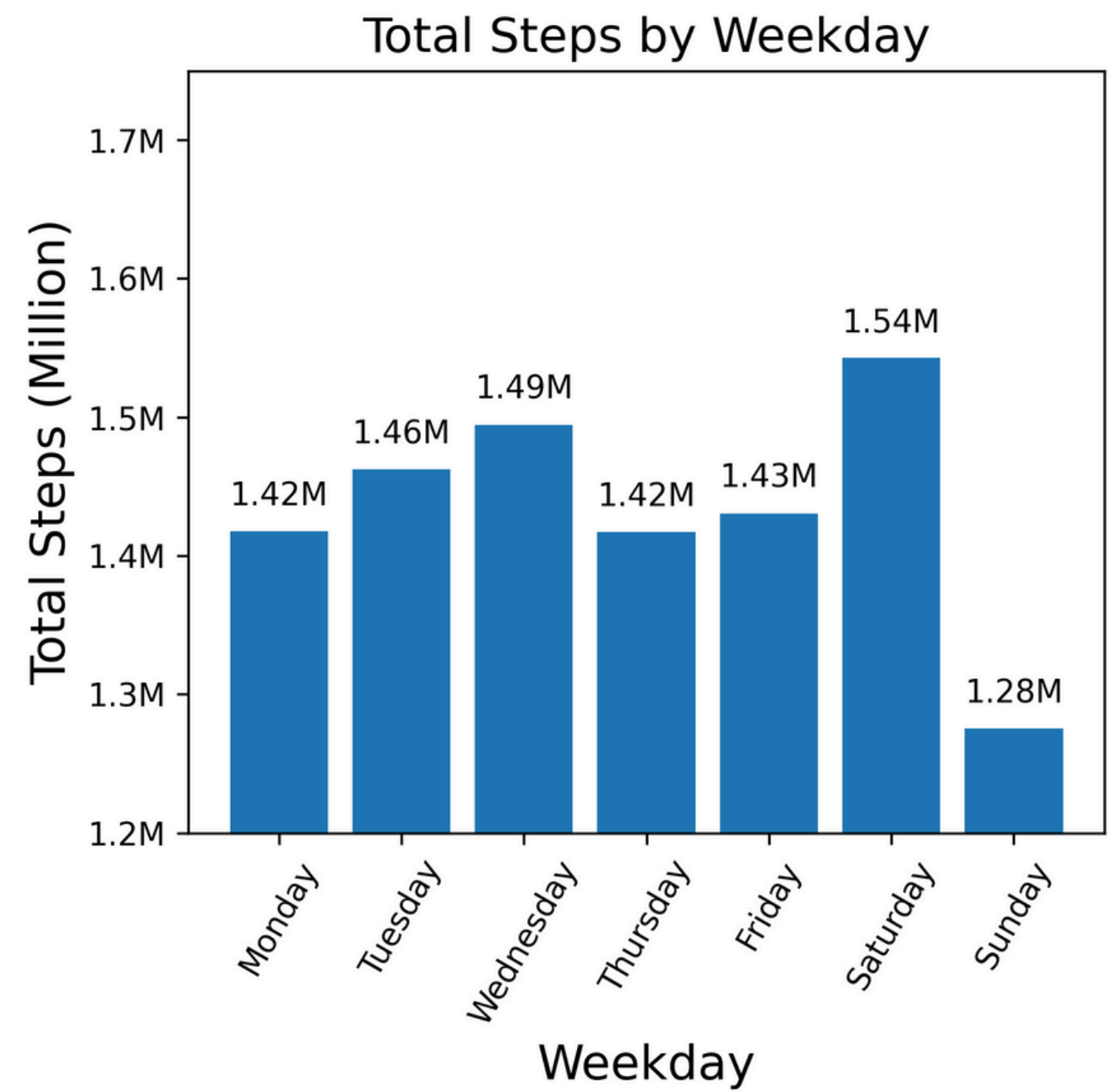


March had lower activity, but calorie levels stayed consistent.



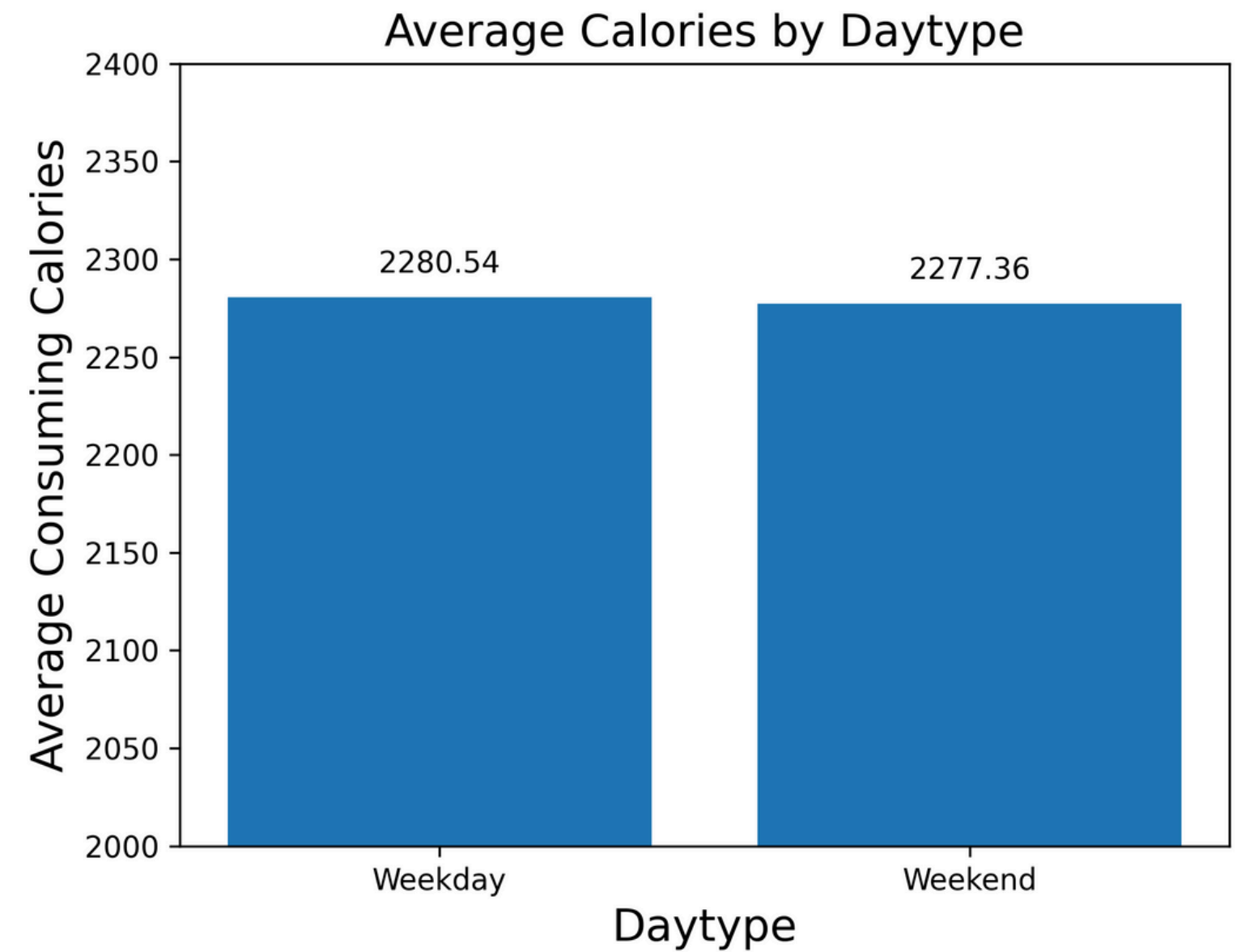
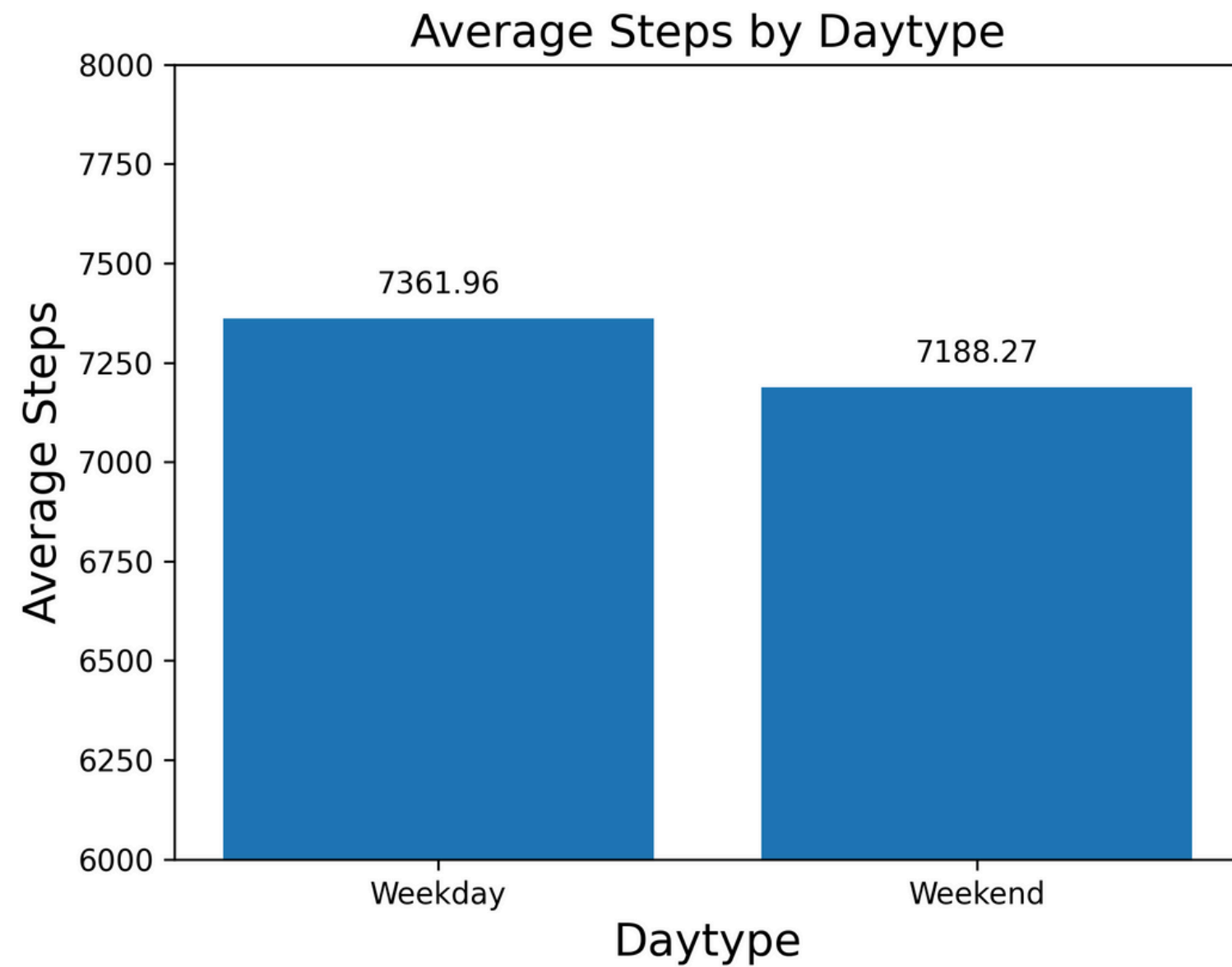
Some non-active users still burned high calories.

Our TractionSteps and Calories by Weekday



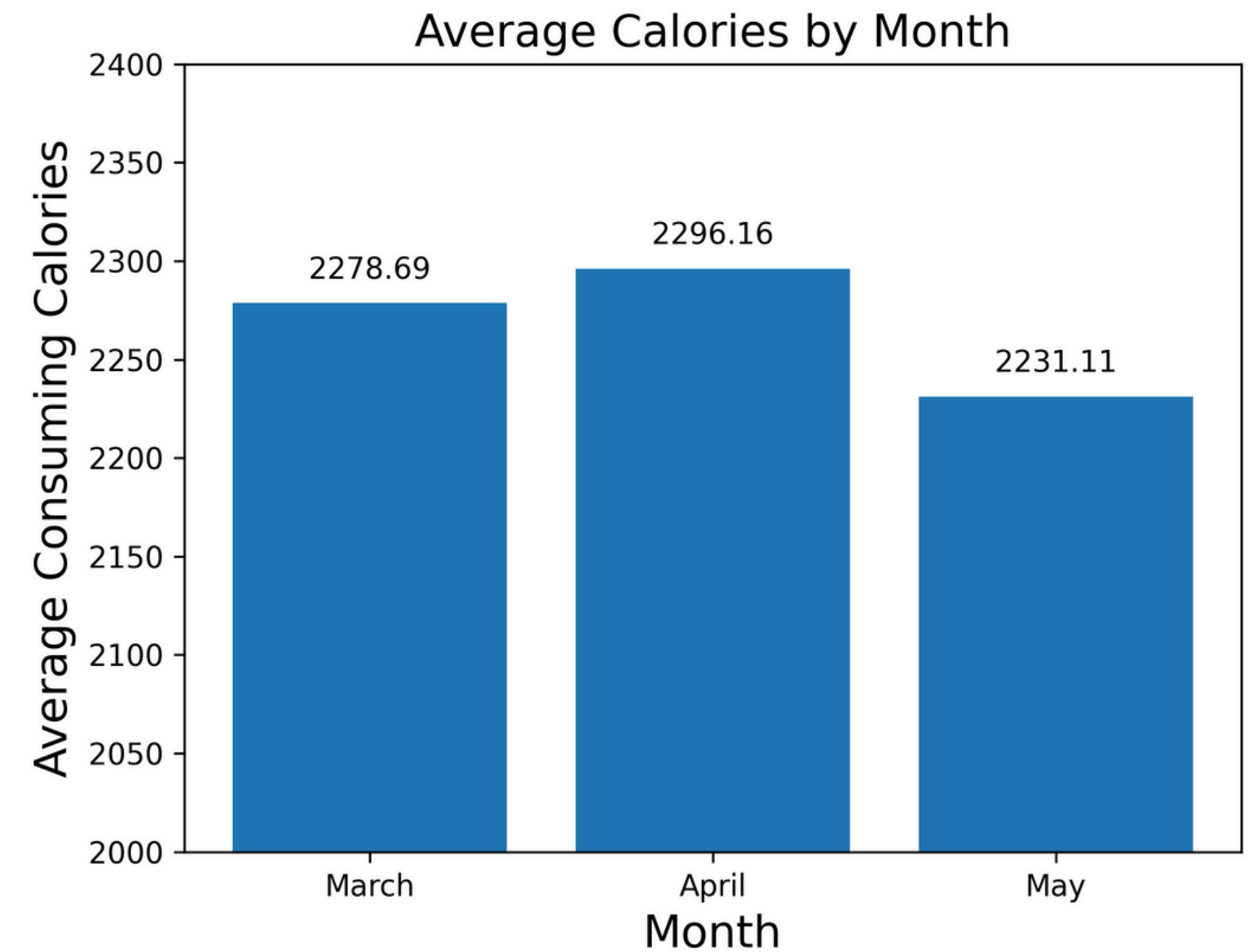
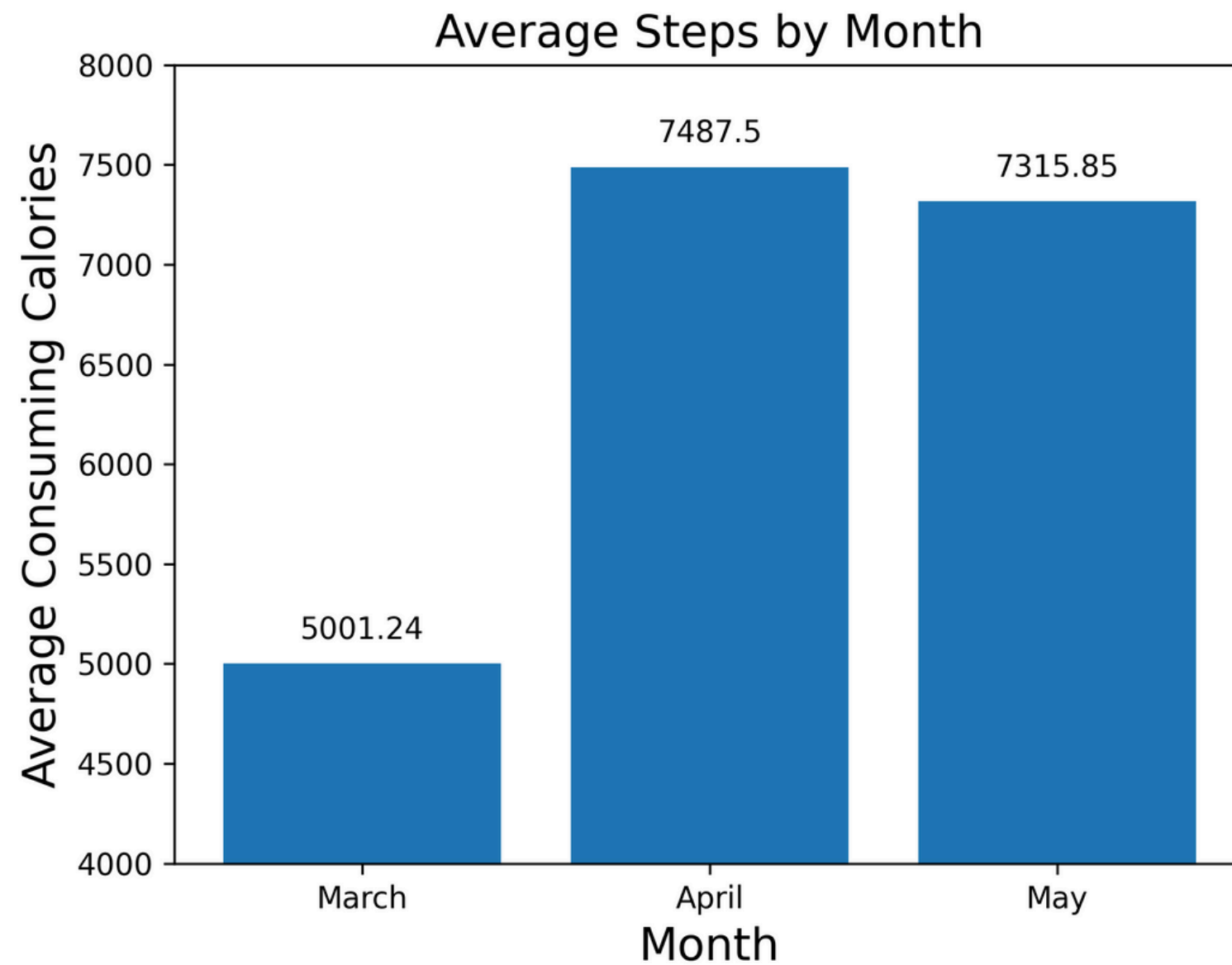
Sunday had the lowest steps, but calories remained steady.

Weektype Comparison



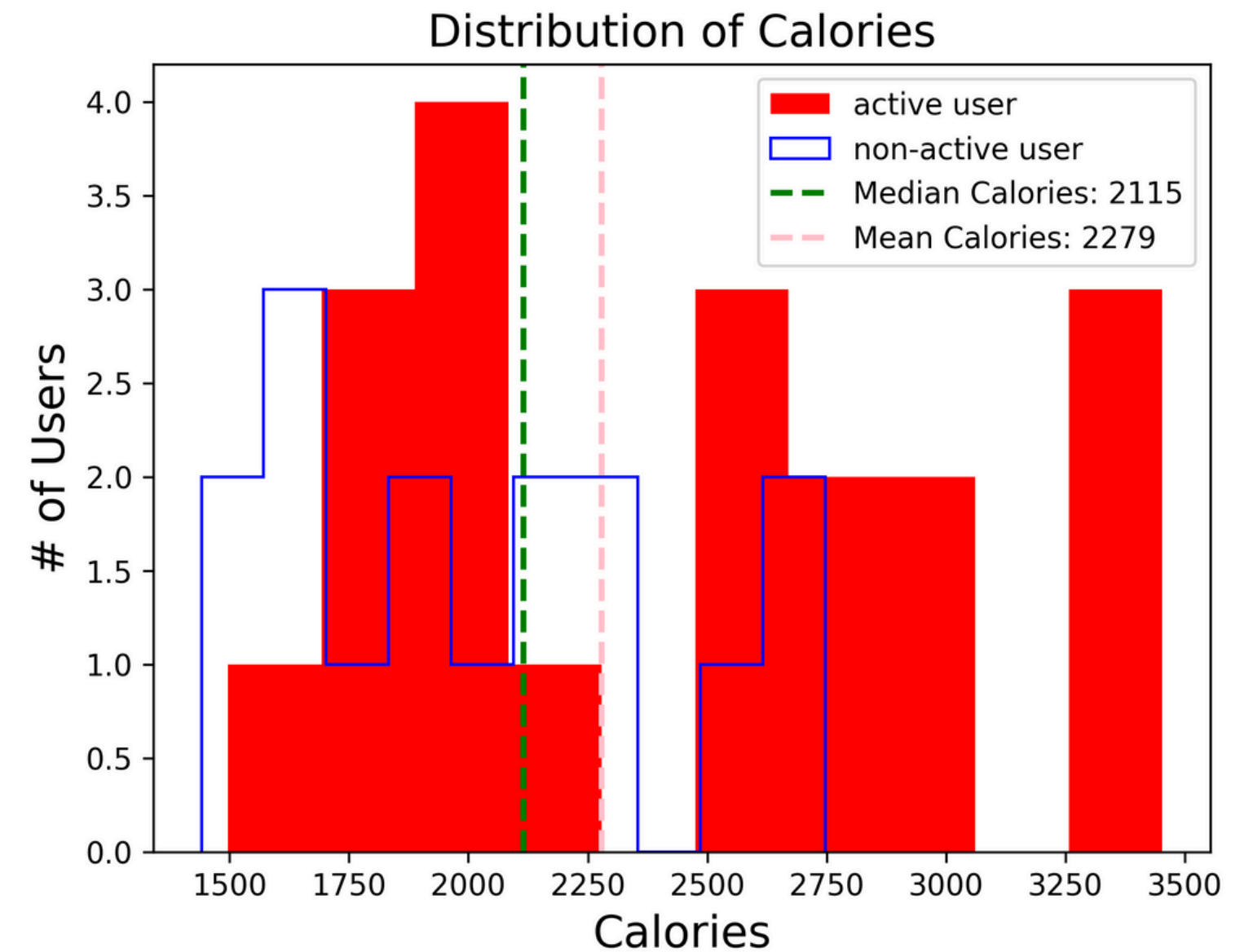
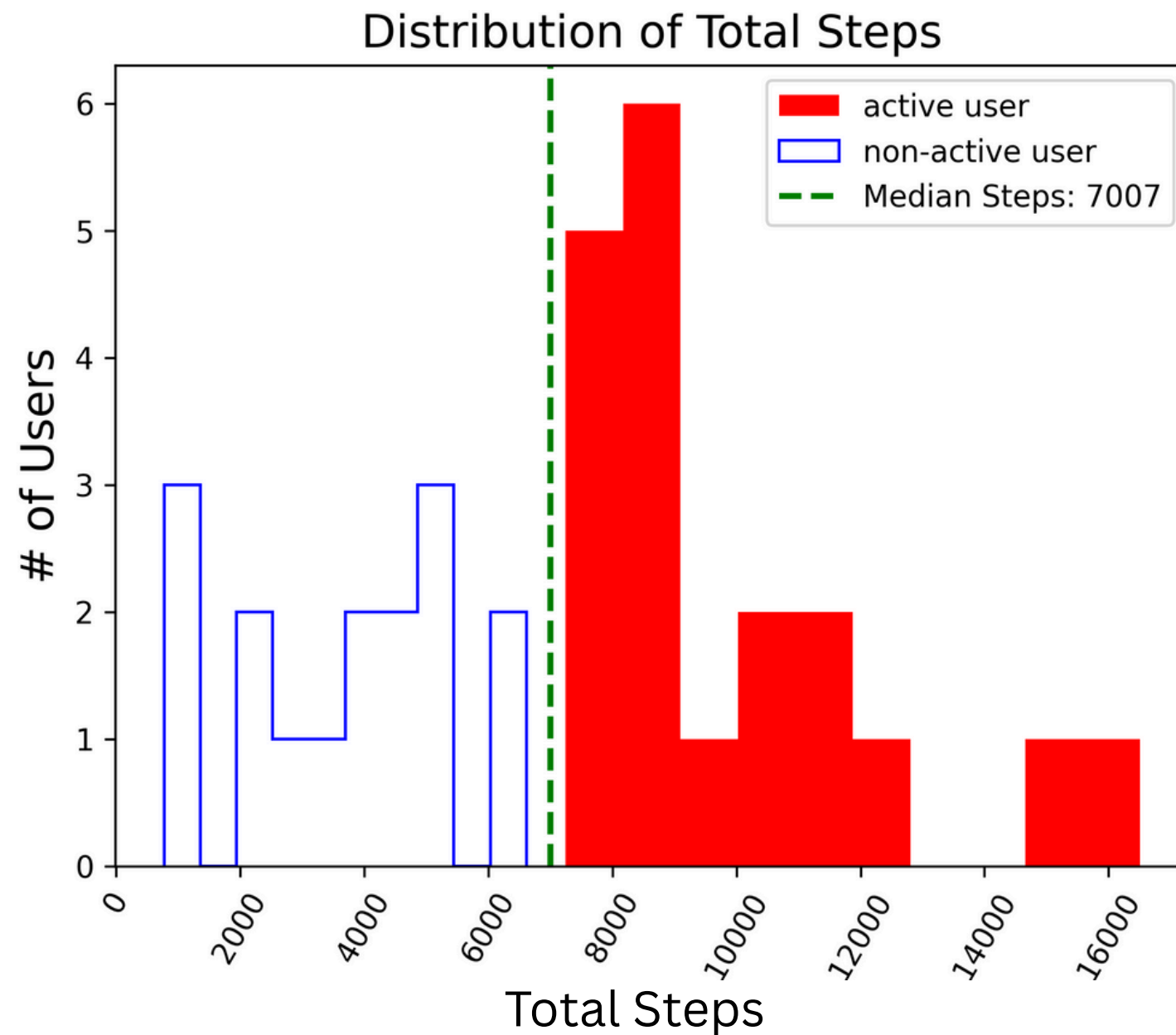
Users are less active on weekends, especially Sundays.

Monthly Trends



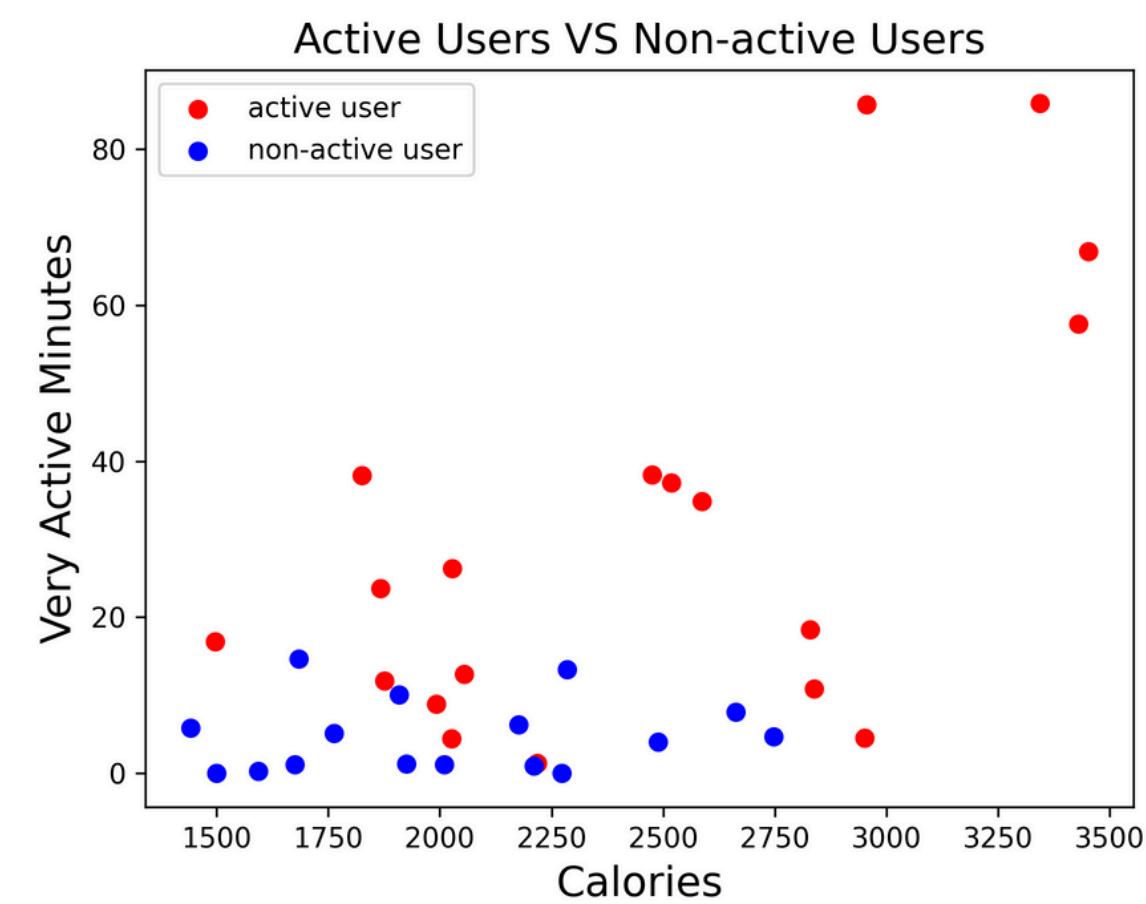
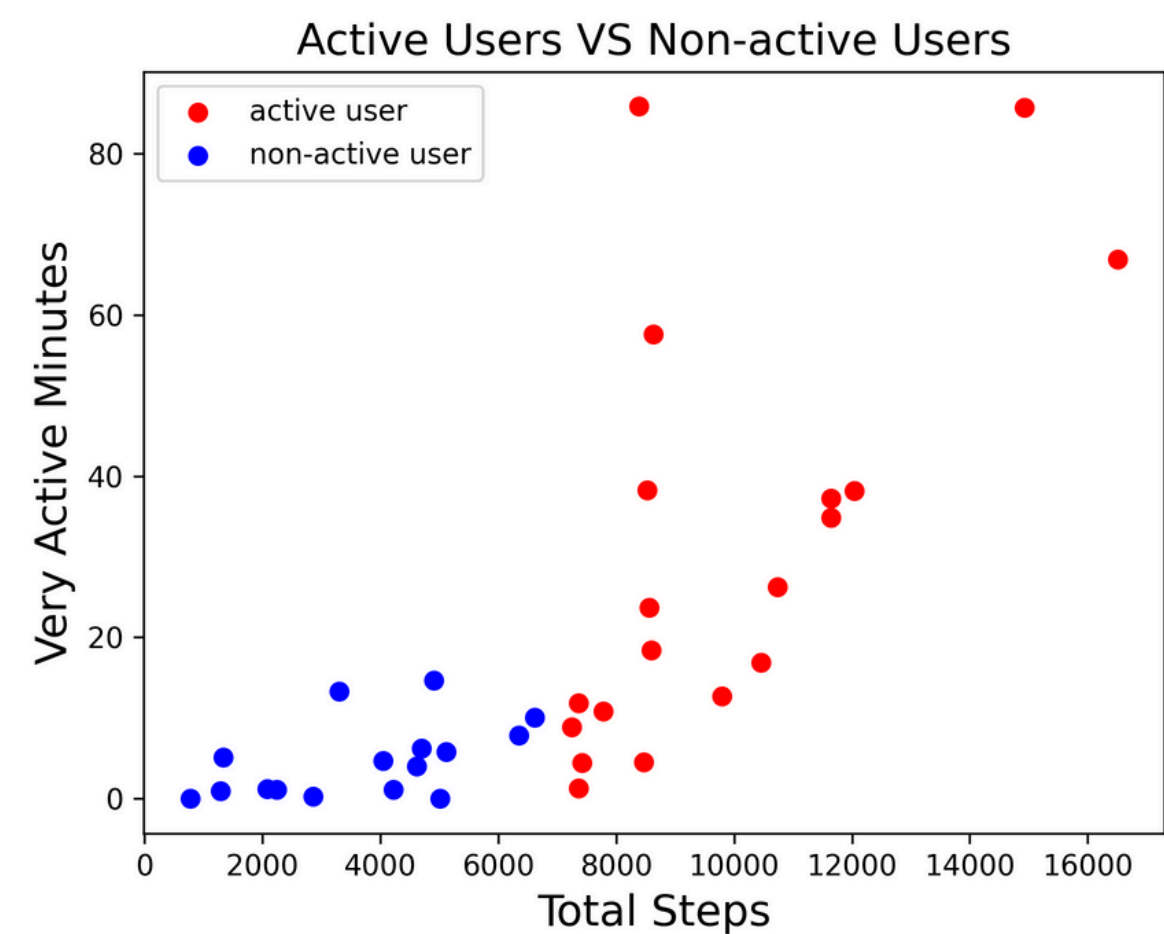
March activity dropped, but calorie burn didn't.

Active vs Non-active Users



Activity level doesn't always align with calorie burn.

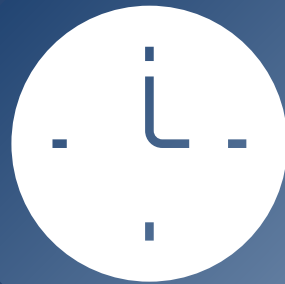
Correlation & Relationships



	VeryActiveDistance	VeryActiveMinutes	Calories	CalculatedTotalDistance
TotalSteps	0.736618	0.675301	0.580400	0.975968

Moderate positive correlation between steps and very active minutes. 8

Recommendations



**Encourage weekend
activity tracking**



**Personalize
recommendations
for non-active users.**



**Calorie analysis
should consider
more than just step
counts.**

Reference

Data Source

This project uses the ****FitBit Fitness Tracker Data**** made available by **Arash Nik** (<https://www.kaggle.com/datasets/arashnic/fitbit>) on Kaggle.

- Dataset title: ***FitBit Fitness Tracker Data***
- Released: 2016
- License: **CC0: Public Domain**
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